HealthGraph: Patient Risk Prediction Using Relation-Aware Graph Attention Networks

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***Abstract*—Thirty-day readmission is a key quality and cost indicator. We present HealthGraph, which models a cohort of diabetic inpatients as a multi-relational graph (clinical similarity, shared diagnoses, shared specialty) and predicts readmission with HealthGAT, a relation-aware GATv2 network implemented from scratch. On the UCI Diabetes-130 dataset we apply leakage-safe feature engineering (out-of-fold target encoding of high-cardinality ICD-9 codes, PCA), tune all models with Optuna, and track experiments in MLflow. Under the published evaluation protocol our pipeline reaches AUROC 0.681, matching or exceeding prior work (0.667). On a strict first-encounter cohort, HealthGAT (AUROC 0.635, ECE 0.005) performs on par with logistic regression and gradient boosting; no difference is significant under DeLong tests. Homophily analysis, relation ablation and a controlled synthetic study, where HealthGAT clearly wins (0.767 against 0.737), show that the graph helps only when its edges carry outcome signal beyond the node features. The model is served through a validated FastAPI/Gradio application that scores unseen patients inductively on their L-hop subgraph, containerised with Docker for Hugging Face Spaces.**

***Index Terms*—graph attention networks, GATv2, hospital readmission, patient similarity graph, diabetes, calibration, Optuna, MLflow**

I. Introduction

Unplanned hospital readmission within 30 days is a widely used quality indicator. It is costly and often preventable with better discharge planning. Diabetic inpatients are a large, high-risk group. Strack et al. [1] released 101,766 diabetic encounters from 130 US hospitals (1999–2008) and found that inpatient HbA1c testing, done in only 18.4% of encounters, was associated with readmission. The data has become the standard public benchmark for readmission modelling. Published models nevertheless plateau at an AUROC of about 0.63–0.69 [2], [3], and clinical scores such as LACE reach about 0.61 on comparable tasks [4], [5].

Almost all published models score each encounter independently. Many of the strongest risk factors, though, are *latent and shared*: a disease phenotype, the practice style of a care team, the burden of a comorbidity cluster. A single record shows only a noisy trace of each, while patients who share diagnoses or a care team pool that evidence. Graph neural networks (GNNs) use exactly this structure: message passing averages noisy evidence over related patients, and attention [7], [8] learns *which* neighbours and *which* relation to trust.

Objectives. (i) Build a reproducible, leakage-free pipeline on a real public dataset, with advanced feature engineering and dimensionality reduction. (ii) Design *HealthGAT*, a relation-aware GATv2 over a multi-relational patient graph, implemented from scratch. (iii) Tune every model with Optuna and track all runs in MLflow. (iv) Benchmark against published results *under their own evaluation protocols*, with significance tests and error analysis. (v) Deploy a validated real-time inference API and a containerised web UI.

Contributions. Beyond the system itself, we contribute (a) a protocol-controlled replication showing how cohort definition, rather than model choice, drives the published numbers; (b) a controlled synthetic study with known ground truth that isolates *when* patient graphs help; and (c) an exact inductive inference scheme that scores a new patient on its L-hop subgraph in milliseconds.

II. Literature Review

Readmission prediction on Diabetes-130. Strack et al. [1] used multivariable logistic regression to study HbA1c testing. Recent machine-learning studies report AUROCs of 0.579–0.667: XGBoost 0.667, LR 0.642, RF 0.630 and a deep network 0.579 with an 80/20 split [2]; and XGBoost 0.664, LightGBM 0.660, LR 0.657 and a stacking ensemble 0.665 with nested cross-validation [3]. Both studies use *all* encounters, so the same patient can appear in training and test folds; [3] acknowledges this explicitly. Gradient-boosted trees [14], [15] are consistently the strongest family, and deep networks have not outperformed them.

Clinical scores and reviews. LACE [5] and similar indices are simple and interpretable but discriminate poorly. A systematic review found most readmission models reach C-statistics of only 0.6–0.7 [6].

Graph learning in healthcare. GCN [9] introduced spectral-inspired neighbourhood averaging, R-GCN [10] added relation-specific weights, and GAT [7] learned neighbour weights with attention. GATv2 [8] showed that GAT's attention is *static* (its neighbour ranking is the same for every query node) and fixed this by applying the non-linearity before the attention vector. In healthcare, GRAM [11] and the Graph Convolutional Transformer [12] model medical-code graphs. MM-STGNN [4] builds an admission-similarity graph and reaches AUROC 0.79 for 30-day readmission on Mayo Clinic and MIMIC-IV data, versus 0.61 for LACE+.

Research gaps. (1) No published GNN study on Diabetes-130 that we found. (2) Benchmarks mix evaluation protocols

with possible patient leakage. (3) GNN papers rarely test whether the graph is *homophilous* for the outcome, although GNN gains depend on it strongly [13]. (4) Few systems address inductive deployment for unseen patients.

III. Methodology

A. Cohort and Preprocessing

The target is $y{=}1$ if `readmitted`="<30". Our main cohort keeps the *first encounter per patient* and removes encounters ending in death or hospice (which cannot be readmitted). This gives 69,973 patients with prevalence 9.0%. Full-batch GAT training on CPU limits the graph experiments to a stratified 20,000-patient subsample; the tabular models are also evaluated on the full cohort (Sec. V-B). Splits are stratified 70/15/15 (train/validation/test) and repeated over three seeds.

Feature engineering (Fig. 1) is fitted on training rows only. *Numeric*: eight utilisation counts, with $\log(1{+}x)$ applied to heavy-tailed prior-visit counts, plus engineered totals (prior visits, number of diabetes drugs, number of dose changes), all z-scored. *Low-cardinality categoricals*: one-hot, with levels seen fewer than 20 times dropped. *High-cardinality*: the admitting specialty (72 levels) and three ICD-9 diagnoses (700+ codes) get log-frequency encoding and smoothed target encoding $\hat{\mu}_k = (\sum_{i\in k} y_i + m\,\bar{y})/(n_k + m)$ with $m{=}30$. This is computed *out-of-fold* (5 folds) for training rows, so no row sees its own label. *Domain grouping*: ICD-9 chapters following [1], as a primary-diagnosis one-hot plus any-position counts. *Codes*: multi-hot of the 100 most frequent 3-character ICD-9 categories. In total this gives 189 input features. Missing values (49% of specialties, and HbA1c not measured in 83% of encounters) are kept as explicit categories, because the absence of a measurement is itself informative.

Dimensionality reduction. PCA on the label-free numeric and one-hot block gives a 16-dimensional *similarity view* (Fig. 4) that keeps 81% of the variance. It is used to build edges, which avoids the curse of dimensionality in k-NN search [25]. t-SNE [25] and UMAP [26] project the learned embeddings for visual analysis.

B. Multi-Relational Patient Graph

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$ with one node per patient and three relations: r_1, clinical similarity (k_s-NN, Euclidean, in the PCA space); r_2, shared diagnoses (k_c-NN, cosine, on ICD multi-hot); and r_3, shared admitting specialty (k_g random peers, a care-team proxy). Edges are symmetrised and deduplicated per relation, and self-loops get their own relation type. Because the edges use features only, the graph cannot leak labels; the setting is transductive (test nodes are present without labels). We measure *outcome homophily* per relation as $P(y_j{=}1 \mid y_i{=}1, (j,i) \in r)$ and compare it with the base rate.

C. HealthGAT

An encoder maps $\mathbf{x}_i$ to $\mathbf{h}_i^{(0)} = \mathrm{ELU}(\mathbf{W}_0\mathbf{x}_i)$. Each relation-aware GATv2 layer computes, per head h,

$$e_{ij}^h = \mathbf{a}_h^\top \,\mathrm{LeakyReLU}\big(\mathbf{W}_s^h\mathbf{h}_j + \mathbf{W}_t^h\mathbf{h}_i + \mathbf{r}_{\tau(j,i)}^h\big), \tag{1}$$

$$\alpha_{ij}^h = \frac{\exp(e_{ij}^h)}{\sum_{k\in\mathcal{N}(i)\cup\{i\}}\exp(e_{ik}^h)}, \quad \mathbf{m}_i = \Big\|_h \sum_j \alpha_{ij}^h \mathbf{W}_s^h\mathbf{h}_j, \tag{2}$$

where $\mathbf{r}_\tau$ is a learned embedding of the edge's relation. It lets each head weigh similarity, diagnosis and specialty neighbours differently, at far fewer parameters than R-GCN's per-relation weight matrices [10]. Applying the non-linearity before $\mathbf{a}_h$ gives dynamic attention [8]. Each layer uses a residual connection, LayerNorm and dropout: $\mathbf{h}^{(\ell+1)} = \mathrm{LN}(\mathbf{h}^{(\ell)} + \mathrm{Drop}(\mathrm{ELU}(\mathbf{m})))$. A jumping-knowledge head maps $[\mathbf{h}^{(0)}\|\mathbf{h}^{(L)}]$ through an MLP to a logit z_i. Training minimises class-weighted binary cross-entropy (positive weight $(1{-}\bar{y})/\bar{y}$) with AdamW, gradient clipping, DropEdge [16] and ReduceLROnPlateau, and stops early on validation AUPRC. The edge softmax is implemented with scatter operations, so the model needs no graph library.

Calibration and decisions. The class weighting shifts all logits upward. Temperature scaling [17] can only rescale them, so we fit Platt scaling [18] $p = \sigma(az + b)$ on the validation split. The operating threshold maximises validation F1 and is applied unchanged to the test set.

D. Hyperparameter Search and Experiment Tracking

Optuna [19] with a TPE sampler maximises validation AUPRC on the seed-0 split. For HealthGAT the search space covers hidden size $\{16, 32\}$, heads $\{2, 4\}$, layers $\{1, 2, 3\}$, dropout $[0.2, 0.6]$, DropEdge $[0, 0.4]$, learning rate $[3{\times}10^{-4}, 5{\times}10^{-3}]$ (log), weight decay $[10^{-4}, 2{\times}10^{-2}]$ (log), and the *graph density* $k_s \in \{5, 10, 20\}$, $k_c \in \{3, 5, 10\}$, $k_g \in \{0, 4, 8\}$, with median pruning after 30 epochs. XGBoost, LightGBM and RF get their own TPE studies; logistic regression uses an exhaustive grid over C. The test split is never used for any selection. Every trial, per-epoch curve, metric and figure is logged to MLflow [20] (SQLite backend) and mirrored to a JSONL log.

E. Evaluation

We report AUROC, AUPRC (the primary metric, given 9% prevalence), log-loss, Brier score, expected calibration error (ECE), F1, sensitivity, specificity, PPV and top-decile capture. All are given as mean±sd over three seeds, with 1,000-sample bootstrap 95% CIs. AUROC differences use the paired DeLong test [21], [22], and decisions at the tuned threshold use McNemar's test. Clinical utility uses decision-curve analysis [23]. Interpretability uses SHAP [24] for the tree baseline, gradient×input for HealthGAT, attention budgets per relation, and a relation ablation.

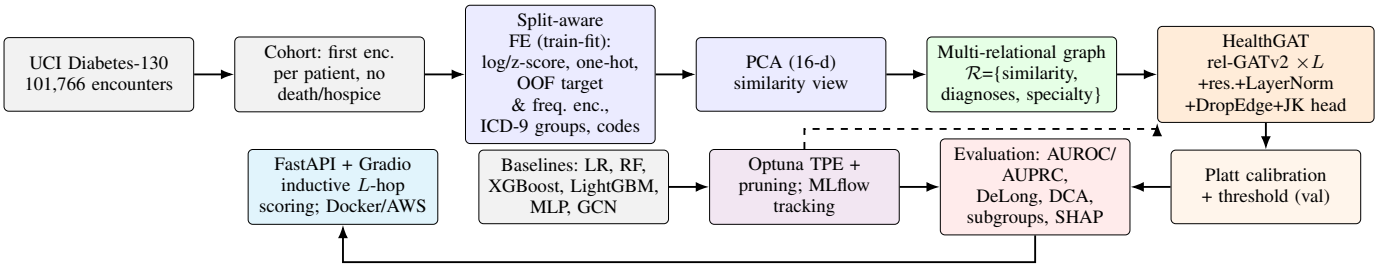


Fig. 1. HealthGraph pipeline. Every fitted statistic (blue) uses training rows only; the same fitted preprocessor and reference graph are reused at inference time.

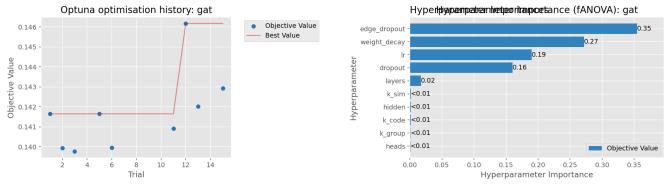


Fig. 2. Optuna search for HealthGAT: optimisation history (left) and fANOVA hyperparameter importance (right).

IV. Implementation and Experiments

Environment. AMD Ryzen 7 7445HS (6 cores / 12 threads), 15 GB RAM, no GPU; Windows 11; Python 3.14, PyTorch 2.14 (CPU), scikit-learn 1.8, XGBoost 3.2, LightGBM 4.6, Optuna 5.0, MLflow 3.17, SHAP 0.52. All models train full-batch on CPU.

Workflow. The pipeline is a Python package driven by YAML configs (`python -m healthgraph all`). Its stages are `tune`, `experiment`, `ablation`, `analyze` and `benchmark`, and each one resumes from checkpoints (Optuna's SQLite store and per-run result files), so interrupted runs continue where they stopped. Each seed re-fits the preprocessor and rebuilds the graph on that seed's training split. Twenty-two pytest tests check the layer mathematics (edge-softmax normalisation, locality of message passing), the exactness of inductive inference, and the API contract.

Tuning outcome. The HealthGAT study ran 16 trials (9 completed, 5 pruned by the median rule, 2 lost to process interruptions). The best configuration (validation AUPRC 0.146) uses 32 hidden units $\times$ 2 heads, 3 layers, dropout 0.46, DropEdge 0.36, learning rate 0.0045 and weight decay 0.0018, with graph density k_s=10, k_c=5, k_g=8 (Fig. 2). MLP and GCN share the tuned GNN configuration, so they differ from HealthGAT only in their aggregation. The tuned graph has 604,990 directed edges over 20,000 patients.

V. Results and Comparative Analysis

A. Discrimination, Calibration and Significance

Table I gives test-set results on the strict cohort. All seven models land in a narrow band, AUROC 0.616–0.644. The best AUROC is logistic regression's (0.644$\pm$0.020), with the MLP (0.642), GCN (0.638) and HealthGAT (0.635$\pm$0.028) close behind. Random forest has the best AUPRC (0.167). The seed-to-seed spread (sd $\approx$0.02–0.03) is as large as the gaps between models: seed 1 is about 0.04 harder for *every* model. The paired DeLong tests (Table II) find no significant difference between HealthGAT and any baseline in any seed, apart from a single borderline p=0.047 against the MLP in seed 2, which is expected by chance across 18 tests. Tuned XGBoost had the highest *validation* AUPRC during tuning, yet the lowest test AUROC (0.616). Tuning on a single split clearly overfits that split, which is a caution for leaderboard-style comparisons.

After Platt scaling every model is well calibrated (ECE $\leq$0.006, Fig. 6). At the validation-tuned threshold HealthGAT flags 49.2% of readmissions with PPV 13.7%, about 1.5$\times$ the base rate, and its top risk decile captures 20.4% of all readmissions. The decision curve (Fig. 12) shows a positive net benefit over "treat all" for threshold probabilities between roughly 5% and 20%, the range relevant to low-cost interventions such as follow-up calls.

B. Comparison with Published Benchmarks

Published numbers are not directly comparable to ours, because the two studies use all encounters with encounter-level splits [2], [3]. We therefore re-ran our tuned tabular models under four protocols (Table III). Under the published protocol (P1, 101,766 encounters) our XGBoost reaches 0.681 and logistic regression 0.672, matching or exceeding the best published results (0.667 and 0.664–0.665). This confirms that the pipeline is competitive. Removing same-patient leakage with a patient-grouped split (P2; under P1 5,047 test patients also appear in training) changes AUROC only slightly (0.680 for XGBoost). The cohort definition matters much more: restricting to first encounters and excluding death and hospice (P3) lowers AUROC to 0.649–0.655. The reason is that later encounters of frequently readmitted patients, which carry rich utilisation history, are removed. Sample size matters too: the 20k-patient graph cohort (P4) costs the tabular models a further 0.01–0.03 against the full 70k (P3). On P4, logistic regression gives 0.644 (identical to Table I, a consistency check) against 0.635 for HealthGAT on the same splits. Cohort definition, not model choice, explains most of the spread in published AUROCs.

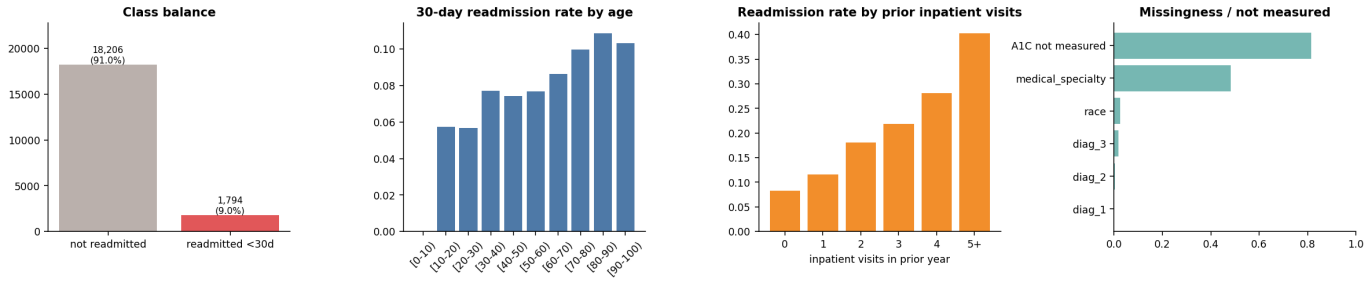


Fig. 3. Exploratory analysis of the 20k-patient cohort: 9.0% prevalence; readmission rises with age and steeply with prior inpatient stays; HbA1c is not measured in most encounters and the specialty is missing for about half.

TABLE I
TEST-SET RESULTS ON UCI DIABETES-130 (STRICT COHORT, 20K PATIENTS; MEAN±SD OVER 3 SEEDS; BEST PER COLUMN IN BOLD).

Model	AUROC	AUPRC	LogLoss	Brier	ECE	F1	Sens.	PPV	Top-10%
Logistic Regression	**0.644±0.020**	0.153±0.019	**0.291**	0.080	0.006	0.222	**0.520**	0.142	0.195
Random Forest	0.633±0.021	**0.167±0.020**	0.291	**0.080**	0.006	0.215	0.491	0.138	0.219
XGBoost	0.616±0.030	0.156±0.015	0.294	0.080	**0.003**	0.202	0.477	0.131	0.197
LightGBM	0.629±0.017	0.159±0.016	0.292	0.080	0.006	0.213	0.457	0.141	**0.219**
MLP (no graph)	0.642±0.019	0.154±0.017	0.292	0.080	0.006	**0.229**	0.489	**0.153**	0.208
GCN	0.638±0.027	0.149±0.018	0.292	0.080	0.006	0.216	0.476	0.140	0.190
HealthGAT (ours)	0.635±0.028	0.151±0.015	0.292	0.080	0.005	0.214	0.492	0.137	0.204

TABLE II
UNCERTAINTY AND PAIRED SIGNIFICANCE TESTS. 95% CIS FROM 1,000 BOOTSTRAP RESAMPLES OF THE SEED-0 TEST SET; ΔAUROC = HEALTHGAT MINUS MODEL (MEAN OVER SEEDS); DELONG p-VALUES PER SEED.

Model	AUROC (seed 0)	AUROC 95% CI	AUPRC 95% CI	ΔAUROC	DeLong p (seeds 0,1,2)
Logistic Regression	0.658	[0.627, 0.692]	[0.141, 0.212]	-0.0085	0.844, 0.249, 0.350
Random Forest	0.650	[0.617, 0.686]	[0.145, 0.218]	0.002	0.601, 0.753, 0.769
XGBoost	0.632	[0.598, 0.667]	[0.130, 0.198]	0.0191	0.072, 0.210, 0.565
LightGBM	0.646	[0.614, 0.682]	[0.145, 0.221]	0.0062	0.269, 0.675, 0.209
MLP (no graph)	0.655	[0.620, 0.690]	[0.143, 0.217]	-0.0066	0.826, 0.111, 0.047
GCN	0.657	[0.624, 0.691]	[0.140, 0.209]	-0.0023	0.868, 0.749, 0.120
HealthGAT	0.657	[0.623, 0.691]	[0.139, 0.209]		

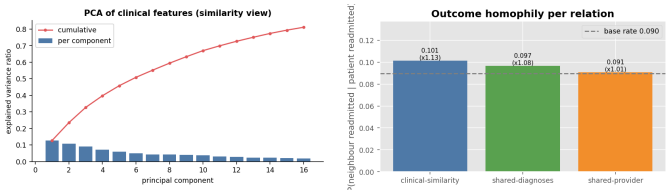


Fig. 4. Left: PCA spectrum of the similarity view. Right: outcome homophily per relation: the probability that a readmitted patient's neighbour is also readmitted, against the 9.0% base rate.

C. Why the Graph Does Not Help Here, and When It Does

Homophily. Fig. 4 (right) shows the outcome signal carried by each relation. A readmitted patient's clinical-similarity neighbour is readmitted 0.101 of the time (1.13× the base rate), a diagnosis neighbour 0.097 (1.08×) and a same-specialty neighbour 0.091 (1.01×). The graph is barely more homophilous than random, and much of the signal it does carry (clinical similarity) is already in each node's own features. Message passing then has little to add, and can even dilute a node's own evidence [13].

Ablation and attention. Retraining HealthGAT without each relation (Table IV, Fig. 8) changes AUROC by less than the seed-to-seed spread (all relations 0.635; self-loops only 0.640). Removing *only* the clinical-similarity edges was slightly worse in all three seeds (0.625), whereas removing every edge was not. This suggests the similarity edges mainly offset noise that the diagnosis and specialty edges introduce, rather than adding signal of their own. The learned attention matches this. HealthGAT gives 48% of its attention to clinical-similarity neighbours, 26% to diagnosis neighbours and 21% to specialty neighbours, and its attention lift towards neighbours who were later readmitted is only 1.02 (1.00 = uniform).

Controlled synthetic study. To check that HealthGAT can exploit a graph when one is informative, we generated 6,000 synthetic patients with eight latent phenotypes and 120 providers whose hidden quality affects the outcome. The provider leaves only noisy traces in individual records (length of stay, polypharmacy, follow-up), and same-provider edges carry 1.54× outcome enrichment. Here the graph models

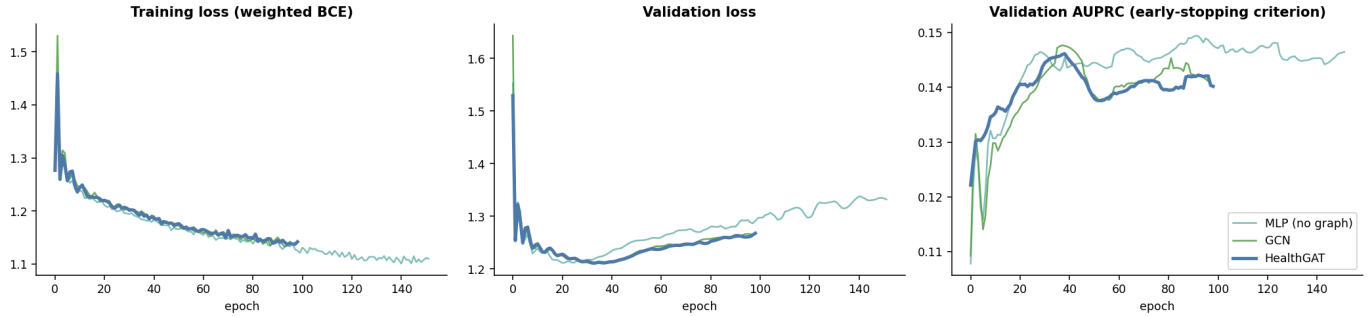


Fig. 5. Training dynamics (seed 0): training loss, validation loss and validation AUPRC (the early-stopping criterion). The graph models converge in fewer epochs than the MLP.

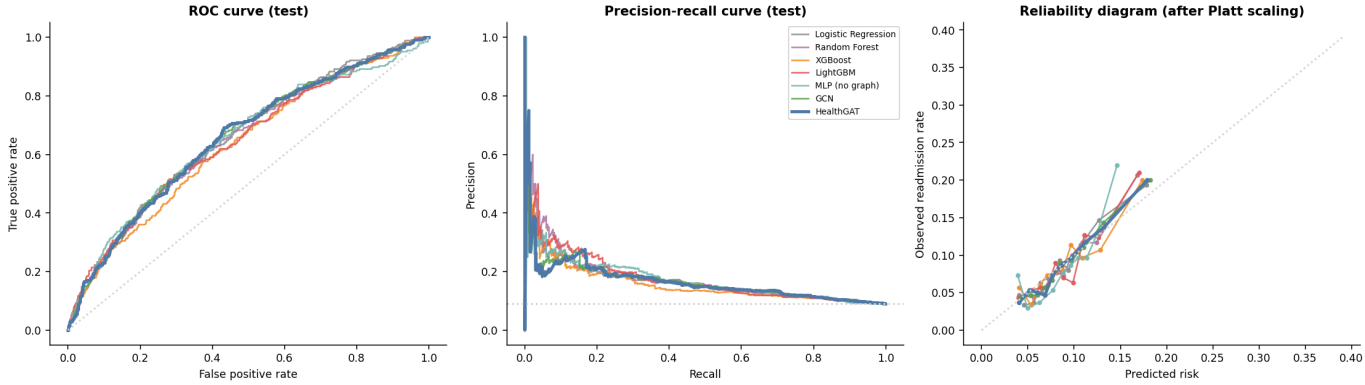


Fig. 6. Seed-0 test set: ROC (left), precision–recall (middle) and reliability diagram after Platt scaling (right). The curves of all models overlap closely.

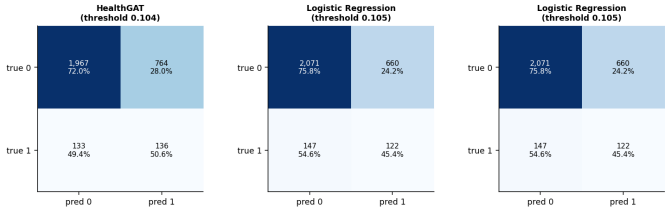


Fig. 7. Confusion matrices at validation-tuned (max-F1) thresholds, seed-0 test set.

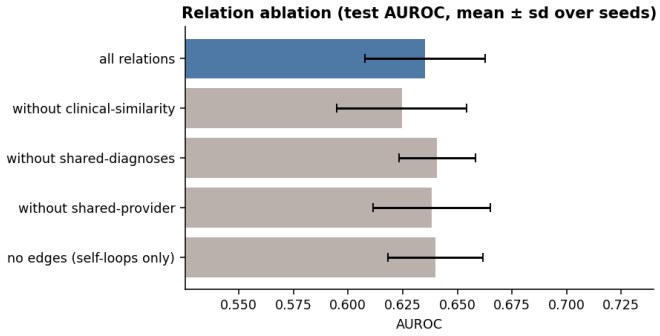


Fig. 8. Relation ablation on UCI (test AUROC, mean±sd over seeds).

clearly beat every tabular model (Table V): HealthGAT reaches 0.767 against 0.737 for the identical network without message passing and 0.729 for LightGBM, with DeLong $p < 0.01$ in two of the three seeds. That closes about a third of the gap to the oracle (0.826), which uses the true risk. Removing provider edges is the most costly ablation (0.767 $\rightarrow$ 0.732), about as costly as removing every edge (0.739), while dropping the other two relations slightly *helps*. HealthGAT and GCN are again within noise of each other (0.767 against 0.763), so in this setting the gain comes from message passing over the right relation more than from attention itself. The architecture therefore works; whether it helps depends on whether the graph carries outcome signal not already in the node features. This can be checked cheaply *before* training, using the homophily diagnostic.

D. Interpretability

SHAP for XGBoost (Fig. 9) and gradient×input for HealthGAT rank the same drivers at the top: prior inpatient stays, discharge home (protective), length of stay, discharge to skilled nursing or rehabilitation, and the engineered diagnosis target and frequency encodings. This confirms the value of the high-cardinality encodings. The embedding projections (Fig. 10) show HealthGAT ordering patients along a risk axis that is not visible in the raw features.

TABLE III
COMPARISON WITH PUBLISHED RESULTS. OUR TUNED MODELS ARE RE-RUN UNDER EACH EVALUATION PROTOCOL (MEAN±SD AUROC, 3 SEEDS). P1 REPRODUCES THE PUBLISHED PROTOCOL.

Source / protocol	Model	AUROC
Emi-Johnson & Nkrumah 2025 [Cureus] [2]	XGBoost	0.667
	Logistic Regression	0.642
	Random Forest	0.630
	Deep neural network	0.579
Salim & Ibrahim 2026 [Healthcare] [3]	XGBoost (tuned)	0.664
	LightGBM	0.660
	Logistic Regression	0.657
	Random Forest	0.650
	Stacking ensemble	0.665
P1: all enc., random split (ours)	Logistic Regression	0.672±0.002
	XGBoost	0.681±0.004
	LightGBM	0.681±0.002
P2: all enc., patient-grouped (ours)	Logistic Regression	0.676±0.009
	XGBoost	0.680±0.005
	LightGBM	0.690±0.010
P3: strict cohort (full) (ours)	Logistic Regression	0.652±0.007
	XGBoost	0.649±0.008
	LightGBM	0.655±0.006
P4: strict cohort (20k) (ours)	Logistic Regression	0.644±0.020
	XGBoost	0.616±0.030
	LightGBM	0.629±0.017

TABLE IV
RELATION ABLATION FOR HEALTHGAT ON UCI (RETRAINED PER VARIANT, 3 SEEDS).

Variant	AUROC	AUPRC
all relations	0.635±0.028	0.151±0.015
without clinical-similarity	0.625±0.030	0.143±0.017
without shared-diagnoses	0.641±0.017	0.153±0.011
without shared-provider	0.638±0.027	0.152±0.014
no edges (self-loops only)	0.640±0.022	0.154±0.013

E. Error Analysis

Subgroups (Fig. 11, test sets pooled over seeds). Calibration holds in every subgroup (largest gap 5.7 percentage points), and discrimination is similar for women and men and for Caucasian and African-American patients. Performance is weakest *within* patients who already have prior inpatient stays (AUROC 0.505 for prior inpatient visits = 2+). Prior utilisation dominates the score, so once it is fixed the remaining features separate risk poorly, and the model under-predicts risk for the most frequent re-admitters.

Error profile. False negatives (410) look like true negatives: they average 0.06 prior inpatient stays, against 0.72 for true positives, with a mean predicted risk of 7.4%. The model mostly misses *first* readmissions that lack any utilisation history. False positives (2,539) are patients with prior stays (0.45 on average) who were not readmitted. Both failure modes point to the same missing information: social factors, post-discharge care and medication adherence, none of which Diabetes-130 records.

TABLE V
CONTROLLED SYNTHETIC STUDY (6,000 PATIENTS, KNOWN GROUND-TRUTH RISK; MEAN±SD OVER 3 SEEDS).

Model	AUROC	AUPRC	Brier	ECE	Top-10%
Logistic Regression	0.733±0.012	0.332±0.010	0.119	0.019	0.246
Random Forest	0.732±0.025	0.312±0.040	0.120	0.025	0.203
XGBoost	0.726±0.022	0.310±0.019	0.121	0.020	0.211
LightGBM	0.729±0.017	0.306±0.016	0.120	**0.017**	0.220
MLP (no graph)	0.737±0.007	0.330±0.005	0.119	0.020	0.244
GCN	0.763±0.007	0.375±0.040	0.116	0.024	0.275
HealthGAT (ours)	**0.767±0.008**	**0.376±0.042**	**0.116**	0.026	**0.280**
Oracle (true risk)	0.826±0.011	0.506±0.043	0.102	0.022	0.344

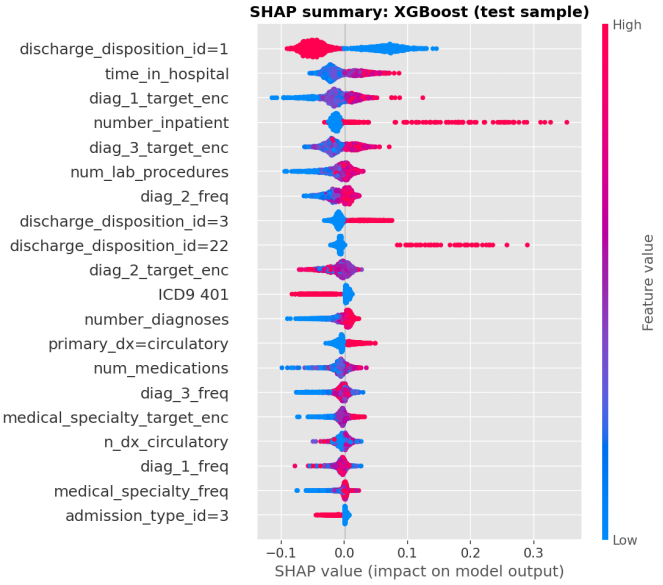


Fig. 9. SHAP summary for XGBoost (2,000 seed-0 test patients).

VI. DEPLOYMENT

Inductive inference. A GNN trained transductively must score patients it has never seen. The deployment bundle (`artifacts/healthgat_uci.pt`, 35 MB) stores the fitted preprocessor, the reference cohort's features and edges, the model weights, the Platt parameters and the decision threshold. A new encounter is preprocessed with the *same* fitted transform and attached to the reference graph with the training rules: k_s nearest neighbours in PCA space, k_c cosine neighbours on its ICD codes, and k_g same-specialty peers drawn with a seed derived from the request, so identical requests get identical edges. An L-layer GNN's output depends only on the node's L-hop in-neighbourhood, so we extract that subgraph from a CSR index and run the forward pass on it alone. A unit test checks that the result equals a full-graph pass on the augmented graph to within 10^{-4}. With the tuned L=3 the subgraph covers about 10k nodes, and scoring takes about 0.3 s per request on CPU (0.5 s with explanations).

Local API. A FastAPI service exposes `POST /predict`, `POST /predict/batch` (at most 100),

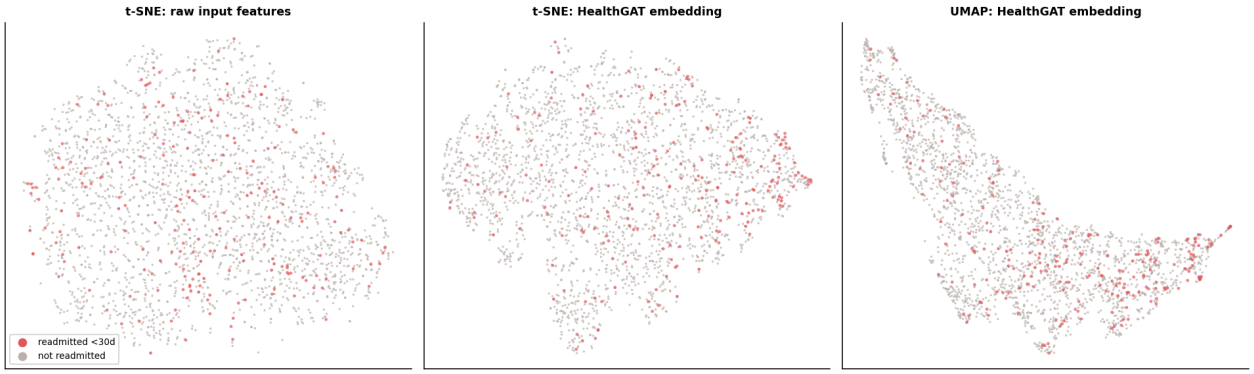


Fig. 10. t-SNE of raw input features against t-SNE and UMAP of the learned HealthGAT representation (2,500 test patients; red = readmitted).

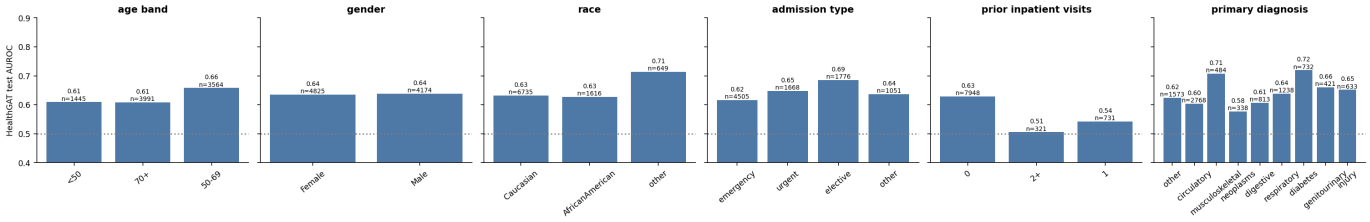


Fig. 11. HealthGAT test AUROC by subgroup (test sets pooled over seeds; n per bar; dotted line = chance).

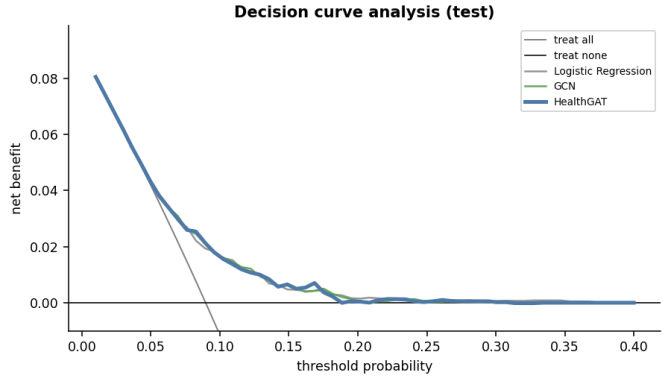


Fig. 12. Decision-curve analysis: net benefit against threshold probability.

`GET /model/info` and `GET /health`, with auto-generated OpenAPI documentation at `/docs`. Pydantic schemas enforce input validation: enumerated age bands and categorical codes, numeric ranges taken from the data dictionary (e.g. length of stay 1–14 days), an ICD-9 format regex, rejection of unknown fields, and a domain rule that refuses patients who died or were discharged to hospice. Invalid requests receive HTTP 422 with a per-field explanation. Each response contains the calibrated risk, a risk band, the follow-up flag, the top gradient-based risk drivers and the most-attended similar past patients, which serves as a case-based explanation. Middleware adds request IDs and timing headers, and logs every call.

Container and cloud. A Gradio UI (Fig. 13) is mounted at `/ui` in the same process. The Docker image (`python:3.12-slim`, CPU PyTorch, pinned dependencies, non-root UID 1000, port 7860, health check; 2.15 GB) was built and tested locally (health check passing, valid predictions, 422 on invalid input). It is deployed on **AWS EC2** (free-tier t3.micro, Amazon Linux 2023, ap-south-1, with an Elastic IP for a fixed address) by one script (`deploy/deploy_aws_ec2.sh`). The script uploads the build context to a private S3 bucket and creates a security group that opens only HTTP. The instance's first-boot script adds 2 GB of swap (the service needs about 0.6 GB of the instance's 1 GB RAM), installs Docker, builds the image and runs it with automatic restart. The live service returns the same prediction as the local container, in about 0.4 s per request. Live UI: http://3.7.153.70/ui; API documentation: http://3.7.153.70/docs. A Hugging Face Spaces configuration is also provided, since Docker Spaces now require a paid plan.

VII. Conclusion and Future Scope

We built HealthGraph, an end-to-end system for 30-day readmission risk built around a relation-aware GATv2 network over a multi-relational patient graph. It covers leakage-safe feature engineering, automated Optuna tuning, MLflow tracking, a protocol-controlled benchmark and a validated, containerised deployment. Under the published evaluation protocol our pipeline matches or exceeds the best reported AUROC on Diabetes-130 (0.681 against 0.667).

The central finding is a nuanced one. On the real data, HealthGAT is as accurate and well calibrated as strong tabular baselines but not significantly better (0.635 against 0.644 for logistic regression). Its patient graph is only weakly outcome-homophilous (1.13$\times$ the base rate at best), and the ablation, the

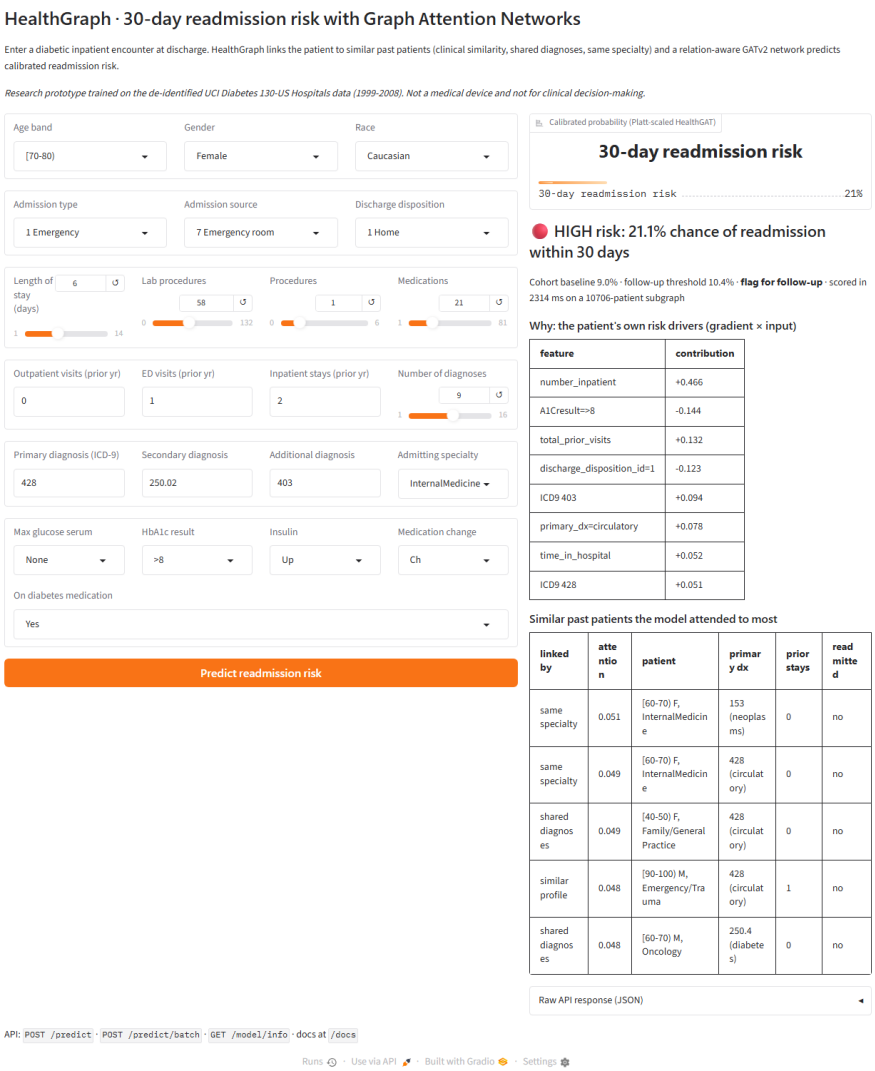


Fig. 13. HealthGraph web UI (Gradio, served at `/ui` by the FastAPI app): the encounter form (left) and the response (right), showing the calibrated risk and band, the patient's own risk drivers, and the most-attended similar past patients with the relation that links them. The nearly uniform attention weights ($\approx$0.05) match the finding that no neighbour stands out on this data.

attention budgets and the controlled synthetic study all tell the same story: patient graphs help when they encode latent shared factors that individual records do not show (on synthetic data 0.767 against 0.737), and add nothing otherwise. Two practical lessons follow. Measure homophily before committing to a GNN, and report results under an explicit cohort definition, because cohort choice moved AUROC more than any model did.

Deployment considerations. The model must be recalibrated, and its reference graph refreshed, for each new site and time period: the data is from 1999–2008, uses ICD-9, and has no post-discharge information. Scoring cost grows with L (the subgraph grows roughly geometrically with depth). Three-layer configurations led the search by only 0.004 validation AUPRC (0.146 against 0.142), so a 2-layer model is a reasonable trade-off when latency matters.

Future scope. (i) Richer, more homophilous relations, such as true provider or ward identifiers, shared medications or temporal encounter chains in MIMIC-IV; (ii) heterophily-aware or label-propagation variants [13]; (iii) mini-batch neighbour sampling to train on the full 70k cohort and beyond; and (iv) prospective validation together with fairness auditing across subgroups.

References

[1] B. Strack, J. P. DeShazo, C. Gennings, J. L. Olmo, S. Ventura, K. J. Cios, and J. N. Clore, "Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records," *BioMed Research International*, vol. 2014, Art. no. 781670, 2014.

[2] O. G. Emi-Johnson and K. J. Nkrumah, "Predicting 30-day hospital readmission in patients with diabetes using machine learning on electronic health record data," *Cureus*, vol. 17, no. 4, Art. no. e82437, 2025.

[3] S. S. Salim and A. A. Ibrahim, "A machine learning approach for predicting 30-day hospital readmission in patients with diabetes," *Healthcare*, vol. 14, no. 9, Art. no. 1185, 2026.

[4] S. Tang, A. Tariq, J. A. Dunnmon, U. Sharma, P. Elugunti, D. L. Rubin, B. N. Patel, and I. Banerjee, "Predicting 30-day all-cause hospital readmission using multimodal spatiotemporal graph neural networks," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 4, pp. 2071–2082, 2023.

[5] C. van Walraven *et al.*, "Derivation and validation of an index to predict early death or unplanned readmission after discharge from hospital to the community," *CMAJ*, vol. 182, no. 6, pp. 551–557, 2010.

[6] D. Kansagara *et al.*, "Risk prediction models for hospital readmission: A systematic review," *JAMA*, vol. 306, no. 15, pp. 1688–1698, 2011.

[7] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. ICLR*, 2018.

[8] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?" in *Proc. ICLR*, 2022.

[9] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. ICLR*, 2017.

[10] M. Schlichtkrull, T. N. Kipf, P. Bloem, R. van den Berg, I. Titov, and M. Welling, "Modeling relational data with graph convolutional networks," in *Proc. ESWC*, 2018, pp. 593–607.

[11] E. Choi, M. T. Bahadori, L. Song, W. F. Stewart, and J. Sun, "GRAM: Graph-based attention model for healthcare representation learning," in *Proc. ACM SIGKDD*, 2017, pp. 787–795.

[12] E. Choi *et al.*, "Learning the graphical structure of electronic health records with graph convolutional transformer," in *Proc. AAAI*, 2020, pp. 606–613.

[13] J. Zhu, Y. Yan, L. Zhao, M. Heimann, L. Akoglu, and D. Koutra, "Beyond homophily in graph neural networks: Current limitations and effective designs," in *Proc. NeurIPS*, 2020.

[14] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM SIGKDD*, 2016, pp. 785–794.

[15] G. Ke *et al.*, "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. NeurIPS*, 2017, pp. 3146–3154.

[16] Y. Rong, W. Huang, T. Xu, and J. Huang, "DropEdge: Towards deep graph convolutional networks on node classification," in *Proc. ICLR*, 2020.

[17] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. ICML*, 2017, pp. 1321–1330.

[18] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*. MIT Press, 1999, pp. 61–74.

[19] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proc. ACM SIGKDD*, 2019, pp. 2623–2631.

[20] M. Zaharia *et al.*, "Accelerating the machine learning lifecycle with MLflow," *IEEE Data Eng. Bull.*, vol. 41, no. 4, pp. 39–45, 2018.

[21] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.

[22] X. Sun and W. Xu, "Fast implementation of DeLong's algorithm for comparing the areas under correlated receiver operating characteristic curves," *IEEE Signal Process. Lett.*, vol. 21, no. 11, pp. 1389–1393, 2014.

[23] A. J. Vickers and E. B. Elkin, "Decision curve analysis: A novel method for evaluating prediction models," *Med. Decis. Making*, vol. 26, no. 6, pp. 565–574, 2006.

[24] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. NeurIPS*, 2017, pp. 4765–4774.

[25] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, 2008.

[26] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform manifold approximation and projection for dimension reduction," arXiv:1802.03426, 2018.